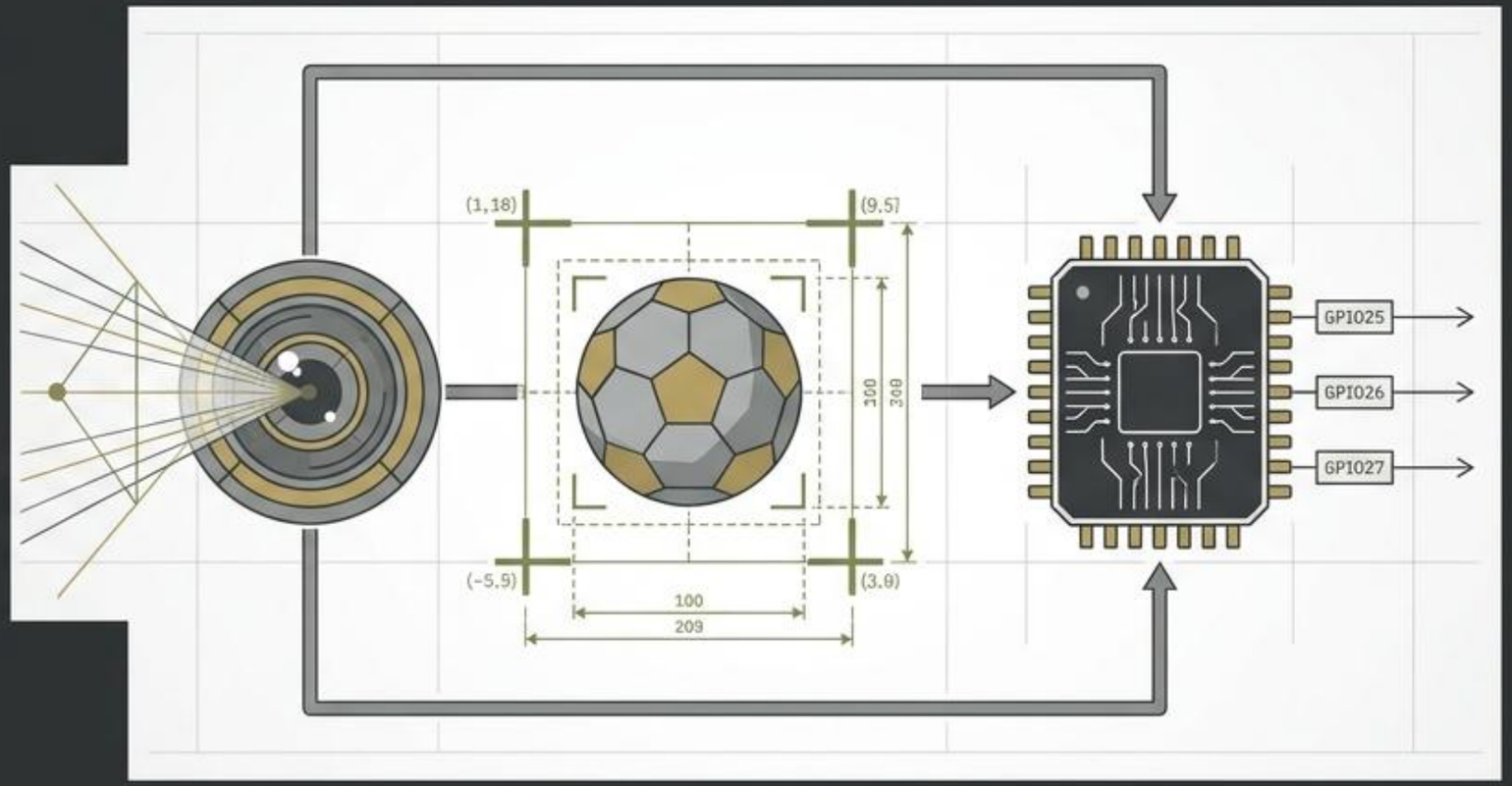
TECHNOLOGY OF ROBOTICS AND IOT CENTRE

AI Vision to Hardware Execution

Practical Work 4: Progressive integration of a YOLO11n object detection pipeline with ESP32 hardware via ROS2 distributed topic architecture.



Gemini Notebook

The Computational Blueprint

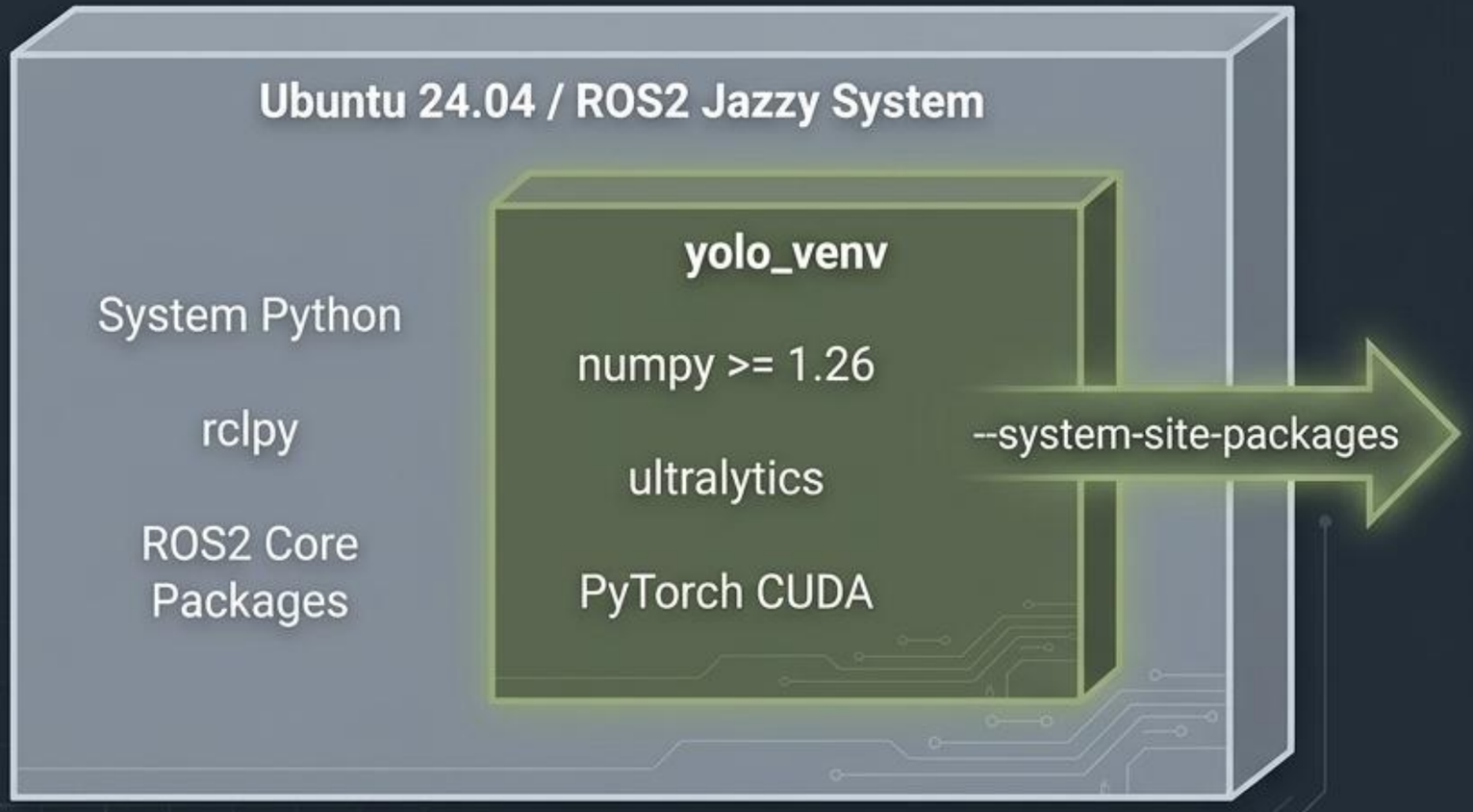
A 6-Phase Pipeline from Raw Pixels to Physical Action



Gemini Notebook

Environment Architecture & Sandboxing

Isolating dependencies while maintaiving system access.




```
python3 -m venv
--system-site-packages
yolo_venv
```

Allows system-level ROS2 packages in, but keeps YOLO dependencies strictly isolated.

Gemini Notebook

Phase 1: Unchained AI Deployment

Validating the general YOLO11n model prior to network integration.

yolo_detection_node.py

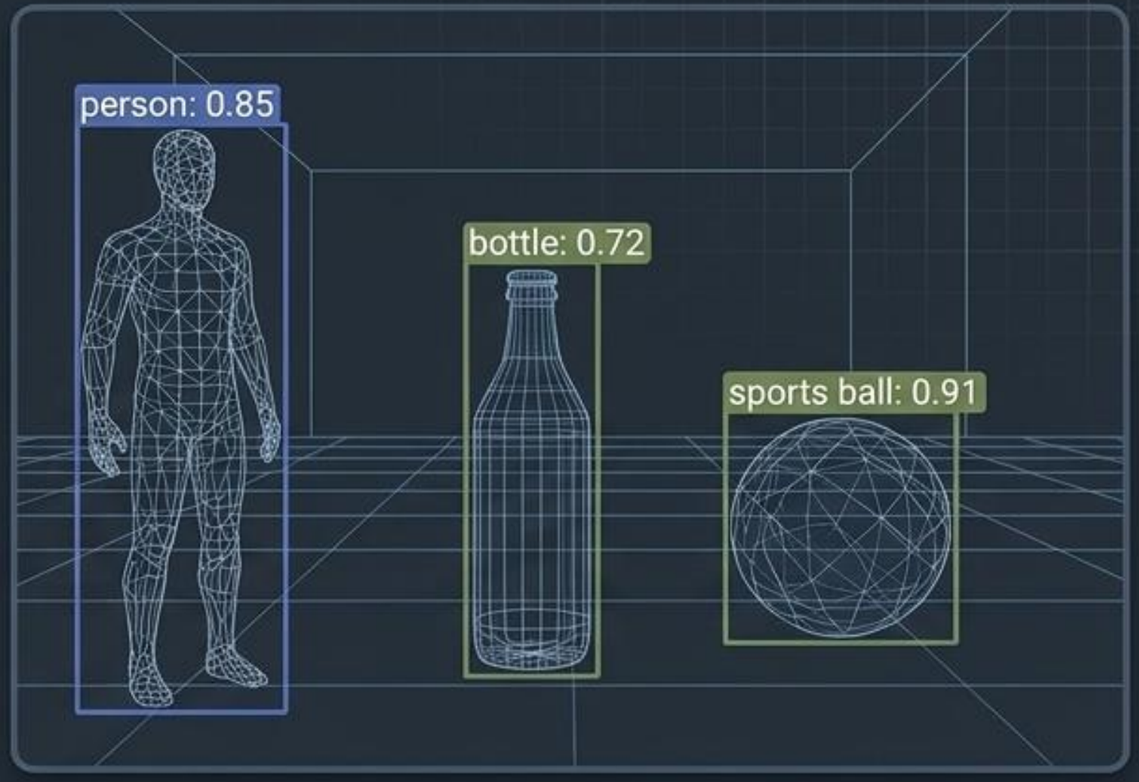
Input: OpenCV V4L2 Video Capture (320x240)

↓

Loop: YOLO11n Predict (conf=0.40)

↓

Output: OpenCV Display Window



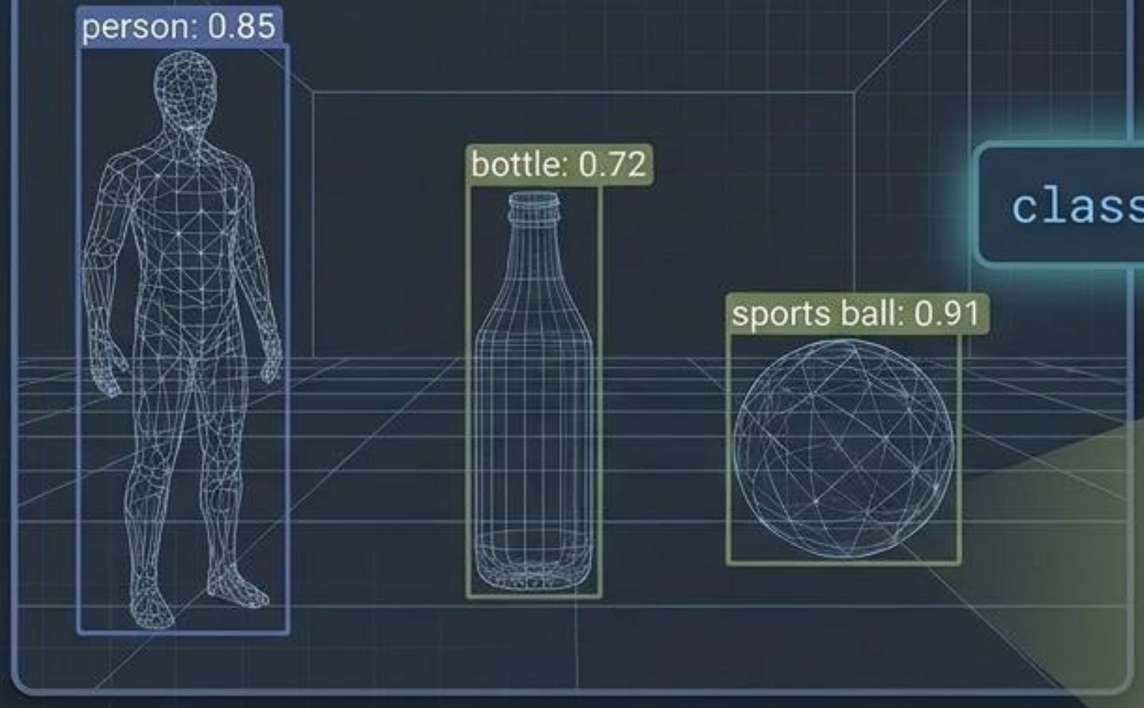
⚠ **Network Isolated:** This node processes video data locally. It does not publish ROS2 topics yet.

Gemini Notebook

Phase 2: Precision Filtering & Geometry

Constraining the model to extract precise spatial coordinates.

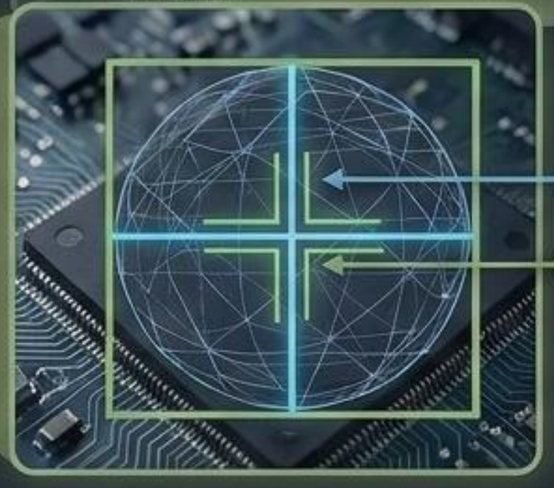
Phase 1: All Classes



`classes=[32]`

Phase 2: COCO Class 32 Only





$$c_x = (x_1 + x_2) \;//\; 2$$

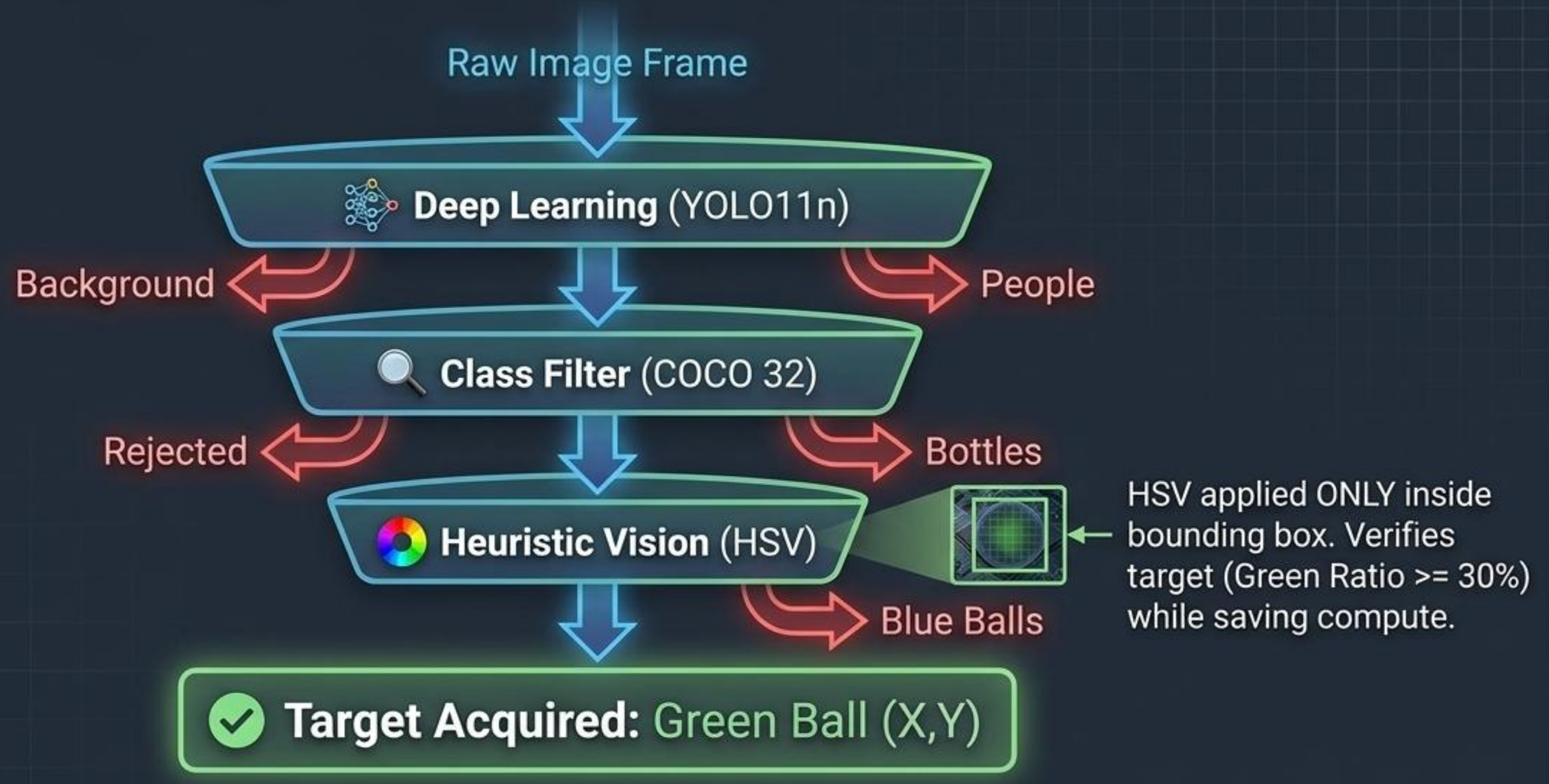
$$c_y = (y_1 + y_2) \;//\; 2$$

Center coordinates extracted for hardware targeting.

Gemini Notebook

The Hybrid YOLO-HSV Funnel

Combining deep learning with traditional computer vision for efficiency.



Gemini Notebook

Phase 3: The AI Meets ROS2

Bridging the vision pipeline into the distributed computational graph.



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Synthesis: Evolution of the Vision Node

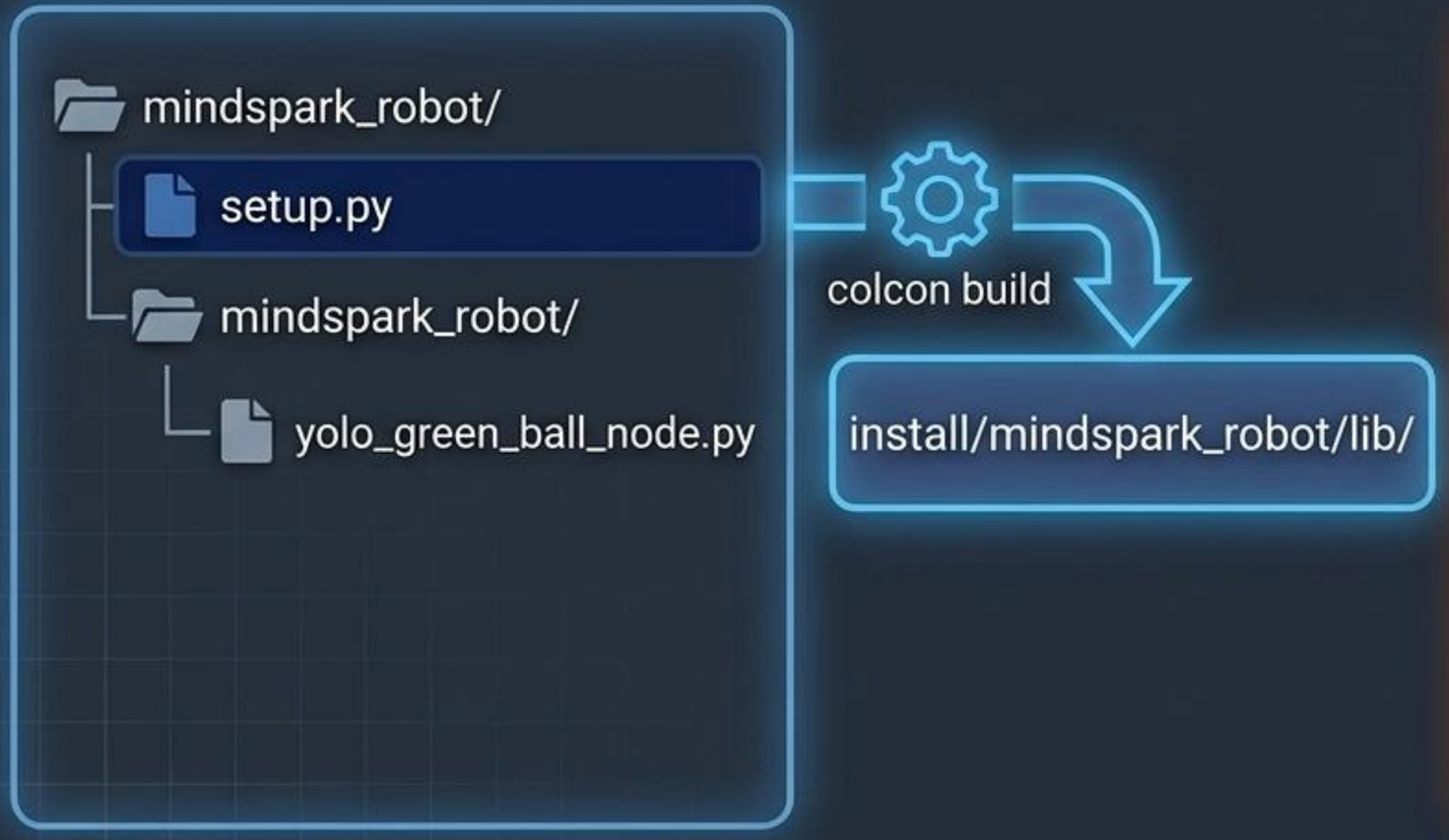
Tracking the progressive refinement of the AI pipeline.

Feature	Phase 1 (General)	Phase 2 (Sports Ball)	Phase 3 (Green Ball)
Target Classes	All 80 COCO Classes	ID 32 Only	ID 32 + Green HSV
Processing Layers	AI Only	AI + Geometry	AI + Geometry + Color
Output Data	Visual Display	Coordinates (cx,cy)	Formatted String
ROS2 Integration	❌ Isolated	❌ Isolated	✅ Fully Integrated

Gemini Notebook

Package Architecture & Registration

Mapping source code to executable binaries.



The Shebang Trap

Registering an entry point generates an executable pointing to the global system Python.

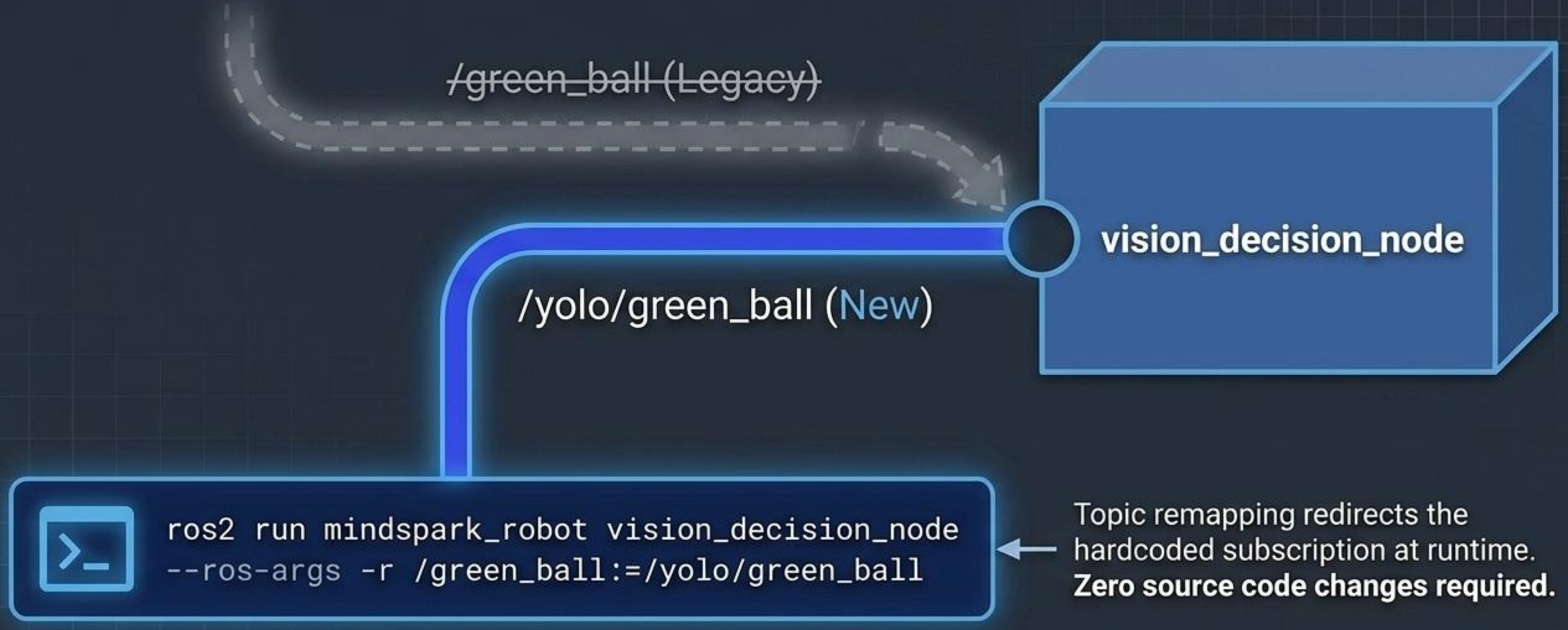
Registration alone does NOT ensure YOLO dependencies are loaded.

The node must be explicitly executed using the isolated yolo_venv interpreter.

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Phase 4: Topic Remapping

Intercepting subscriptions to reuse legacy code.

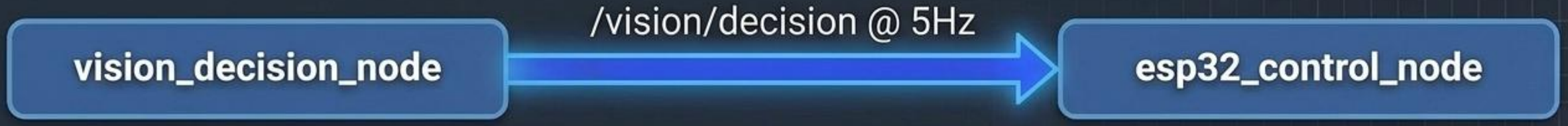


Topic remapping redirects the hardcoded subscription at runtime. **Zero source code changes required.**

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Phase 5: Hardware Control Logic

Translating spatial coordinates into microcontroller pin activations.

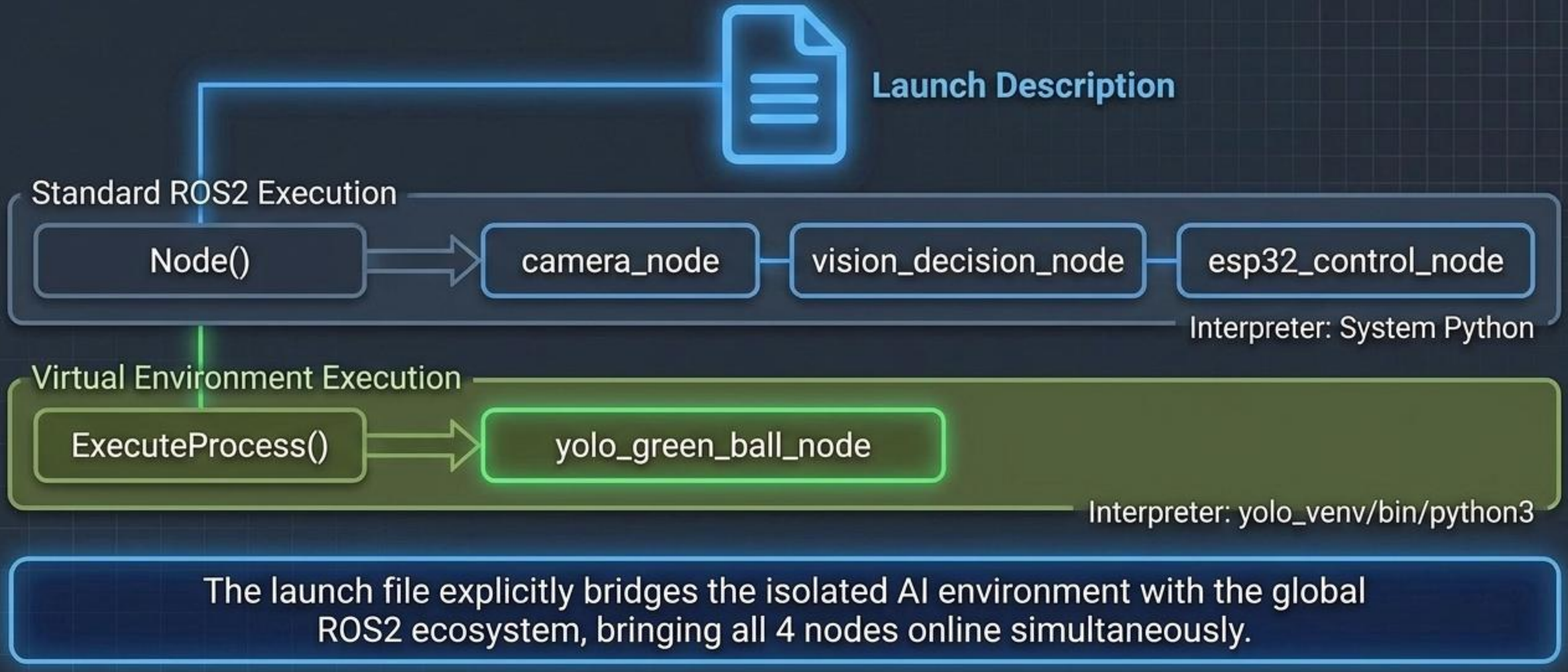


String Payload	Computed Position	ESP32 Pin	LED Output
DETECTED: x < threshold	L (Left)	GPIO25	Red
x in center zone	C (Center)	GPIO26	Yellow
x > threshold	R (Right)	GPIO27	Green
NOT DETECTED	N (None)	N/A	ALL OFF

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Phase 6: Launch Automation

Orchestrating the system via pw4_yolo_hardware.launch.py

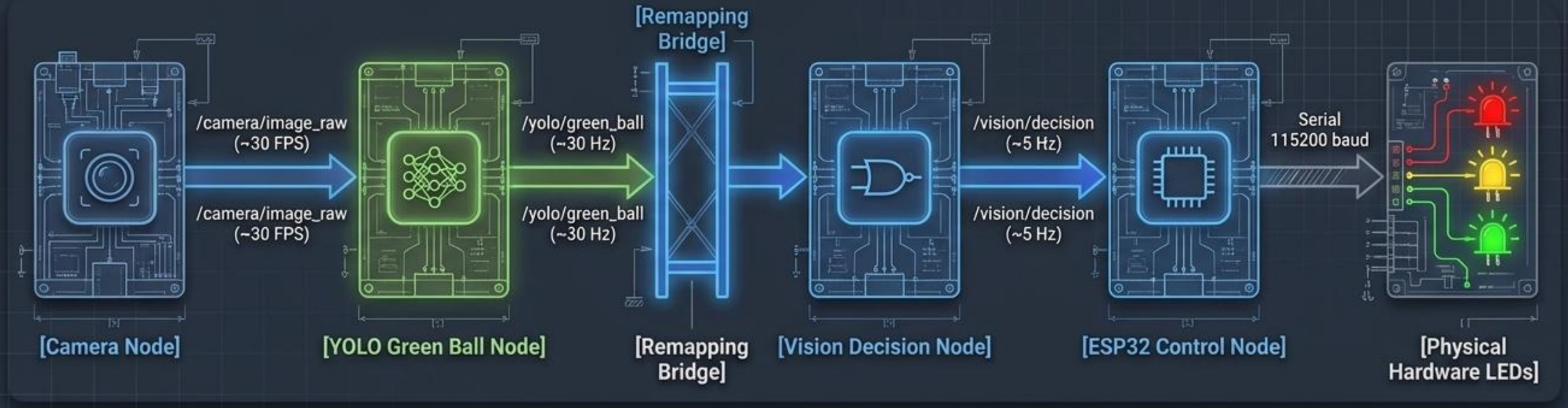


The launch file explicitly bridges the isolated AI environment with the global ROS2 ecosystem, bringing all 4 nodes online simultaneously.

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The Complete Computational Graph

End-to-end data flow: Raw pixels to physical hardware.

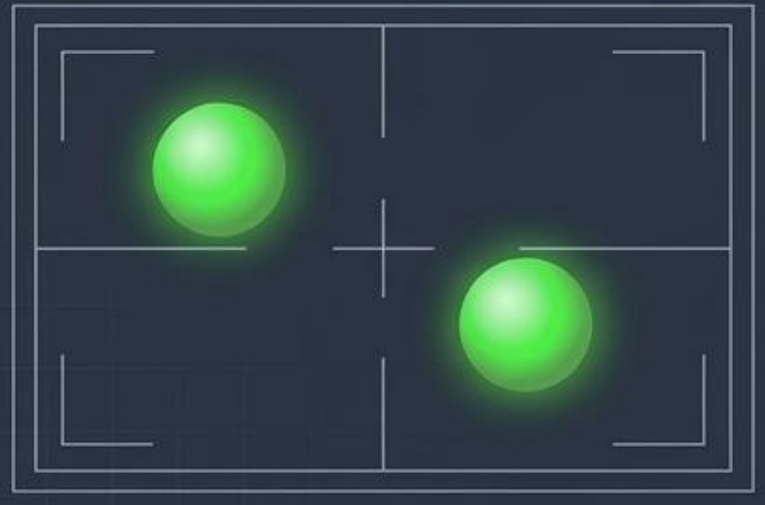


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Practical Verification & Edge Cases

Ensuring system resilience and hardware safety.

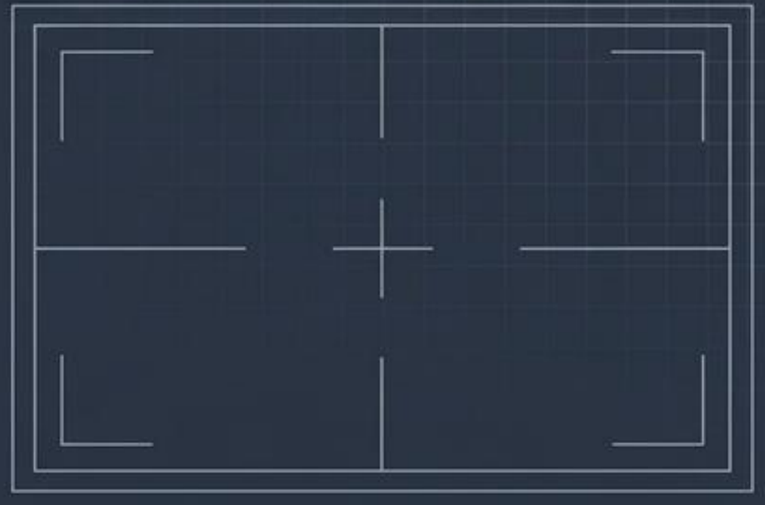
Edge Case: Multiple Targets



Condition: Two valid targets enter the frame.

Solution: Node sorts array by best_conf. The AI locks onto the highest confidence target and ignores secondary targets, preventing coordinate ping-ponging.

Edge Case: Target Lost



Condition: The target leaves the frame or is occluded.

Solution: AI publishes 'NOT DETECTED'. Decision node translates this to 'N'. ESP32 turns all LEDs OFF, ensuring hardware does not lock into its last known state.

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Engineering Takeaways

Core architectural principles for robotics integration.

AI Integration Strategy

Avoid relying solely on deep learning.
Combine pretrained models (YOLO11n) with fast heuristics (HSV) to achieve highly efficient, precise targeting while conserving compute.

Architecture Modularity

Never rewrite functioning legacy code.
Utilize ROS2 topic remapping to seamlessly bridge new sensor pipelines into existing decision logic at runtime.

Deployment Orchestration

Use virtual environments to strictly secure dependencies.

Design launch files that explicitly bridge isolated interpreters with the global ROS2 runtime ecosystem.

```
End of Practical Work 4.
Domain ID: 30
System Status: Online.
```

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